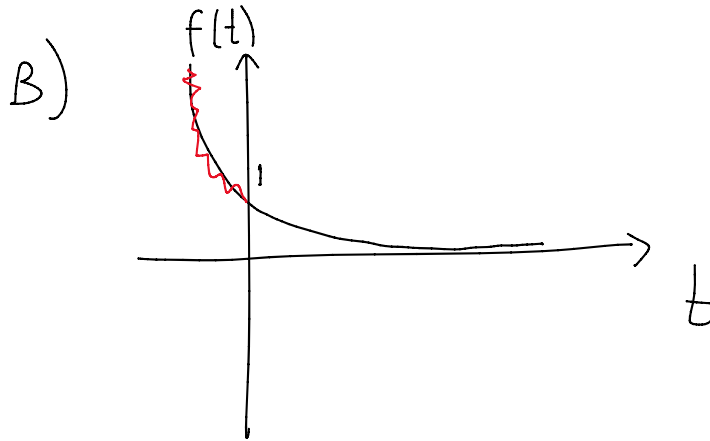


☒ Ejercicio nº 11:

Halle las Transformadas de Fourier de las siguientes funciones y dibuje su espectro continuo de amplitud:

a) $f(t) = \begin{cases} 1 & \text{si } |t| < 3 \\ 0 & \text{si } |t| > 3 \end{cases}$

b) $f(t) = \begin{cases} e^{-t} & \text{si } t > 0 \\ 0 & \text{si } t < 0 \end{cases}$



$$F[f(t)] = F(\omega) = \int_{-\infty}^{\infty} f(t) \cdot e^{-j\omega t} dt$$

$$F(\omega) = \int_0^{\infty} e^{-t} \cdot e^{-j\omega t} dt$$

$$= \int_0^{\infty} e^{-t(1+j\omega)} dt$$

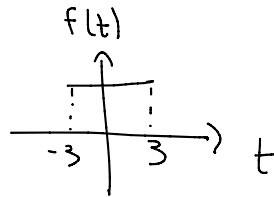
$$= \left. \frac{e^{-t(1+j\omega)}}{-1-j\omega} \right|_0^{\infty}$$

$$= \frac{0 - 1}{-1-j\omega}$$

$$= \frac{1}{1+j\omega}$$

$$= \left[\frac{1}{1+j\omega} \right]^0$$

$$A) f(t) = \begin{cases} 1 & \text{si } |t| < 3 \\ 0 & \text{si } |t| > 3 \end{cases}$$



$$F(\omega) = \int_{-3}^3 1 \cdot e^{-j\omega t} dt$$

$$= \frac{e^{-j\omega t}}{-j\omega} \Big|_{-3}^3$$

$$= \frac{-1}{j\omega} (e^{-3j\omega} - e^{3j\omega})$$

$$= \frac{1}{\omega} \left(\frac{e^{3j\omega} - e^{-3j\omega}}{2 \cdot j} \right) \cdot 2$$

$$= \frac{2}{\omega} \cdot \sin(3\omega) \quad (\text{si } \omega \neq 0)$$



Ejercicio n° 12:

En base al ejercicio anterior (parte a), calcule el valor de la integral:

$$\int_0^{\infty} \frac{\sin(x)}{x} dx$$

Def. Antitransformada de Fourier:

$$f(t) = F^{-1}[F(\omega)] = \frac{1}{2\pi} \int_{-\infty}^{\infty} F(\omega) \cdot e^{j\omega t} d\omega$$

$$f(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} \frac{2}{\omega} \cdot \text{sen}(3\omega) \cdot e^{j\omega t} d\omega$$

No lo
queremos
ahí

$$F(0) = 1 \quad (\text{Gráfico})$$

$$f(0) = \frac{1}{2\pi} \int_{-\infty}^{\infty} \frac{2 \cdot \text{sen}(3\omega)}{\omega} d\omega$$

$$\frac{1}{\pi} \int_{-\infty}^{\infty} \frac{\text{sen}(3\omega)}{\omega} d\omega = 1$$

Cambio de
variable:

$$3\omega = x$$

$$\omega = x/3$$

$$d\omega = dx/3$$

$$\frac{1}{\pi} \int_{-\infty}^{\infty} \frac{\text{sen}(x)}{x/3} \frac{dx}{3} = 1$$

$$\frac{1}{\pi} \int_{-\infty}^{\infty} \frac{\sin(x)}{x} = 1$$

$$\int_{-\infty}^{\infty} \frac{\sin(x)}{x} = \pi$$

$$\int_0^{\infty} \frac{\sin(x)}{x} = \frac{\pi}{2}$$

Porque $f(x)$ es par

